

Arastun Mammadli

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Profile

An MSc student in Applied Computational Science at Imperial College London focused on practical applications of machine learning, deep learning, and large language models. Interested in scientific machine learning and AI agent orchestrations. My thesis work is on Azure AI Services @ Microsoft, applying emerging agentic frameworks on campus energy management.

Education

Imperial College London, MSc Applied Computational Science and Engineering Sept 2024 – Sept 2025

- Term 1: A (predicted)
- Modules include: Computational Mathematics, Deep Learning, Data Science & Machine Learning

University of Exeter, BSc Computer Science Sept 2020 – May 2024

- First-Class Honours, Dean's list for Year 1 (top 5%)

Dunya IB School, International Baccalaureate Diploma Sept 2018 – May 2020

- Grade: 40, Extended Essay: Genetic Algorithms in Constrained Optimization Problems

Experience

Microsoft, Research Intern June – Sept 2025

- Worked on conversational multi-agent workflows for campus energy management. Developed a full-stack prototype using Azure AI Agent Services, Azure AI Search (RAG), and Azure AI Language (NLP services).
- Used Azure Evaluation SDK for model selection, prompting, query complexity and routing ablations.
- Independently managed project timelines, and weekly communicated progress to Microsoft stakeholders.

Imperial College London, Internal Project May 2025

- Developed discrete and continuous genetic algorithm solvers to optimise circuit design.
- Developed circuit simulator (mass-balance solver) and a validity checker for performance evaluations.
- Collaborated with team of 8, managed the team's GitHub repository for software sustainability.

Imperial College London, Internal Project Jan 2025

- Developed Deep-Learning (U-Net, Conv-LSTM) models for real-time lightning storm prediction.
- Strengthened expertise in data and time-series analysis, deep learning, and scientific computing.

Embawood LLC, Software Developer Intern, Baku, Azerbaijan May – Aug 2023

- Developed and maintained Java-based applications using the Spring Framework.
- Worked with engineering teams on Spring Boot RESTful APIs for internal tool support, participating in code reviews and technical presentations.
- Wrote unit and integration tests with JUnit to reduce production bugs.

Skills

Programming Languages: Python, C++, C, Java, C#

Tools & Frameworks: Git, REST APIs, Azure AI Services, Azure Foundry, React, PyTorch, MPI

Languages: Fluent in English, Azerbaijani, Russian, and Turkish, Limited working in Spanish

References available on request